Utilizing Ontologies in Comparative Urban History Research: A Geospatial Analysis

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Introduction

The processing of geodata and historical location data in the field of digital history has, thus far, been predominantly conducted with the assistance of geographic information systems (GIS), primarily for the purpose of data visualisation (Bol, 2011). As part of the HisMaComp project (Historical Survey Maps and the Comparative Study of the Functionality and Morphology of Urban Space), a data-driven approach to analyse the historical and spatial-topographical development of towns in a comparative manner is employed.

The research question is: How can the topography, functionality and morphology of historical urban spaces be recorded and analysed comparatively over time? GIS are insufficient for analysing all these aspects together. However, by combining GIS and semantic web technologies, the complexity of the information to be processed can be increased, thereby enabling in-depth and multidimensional comparative analyses.

Context of the project

The German-Polish collaborative research project HiSMaComp is carried out by two research teams at the Institute for Comparative Urban History (IStG) in Münster and the Uniwersytet Mikołaja Kopernika in Toruń. It is funded by the BEETHOVEN 4 programme of the DFG and NCN. Six towns in Poland and Germany serve as case studies based on three town types: Warsaw and Magdeburg as 'pre-modern metropolises', Olsztyn/Allenstein and Ochsenfurt as 'medieval collegiate towns', and Inowrocław/Hohensalza and Bad Pyrmont as 'modern spa towns' (Szende and Szilágyi, 2019).

The core of the comparative analysis of the case studies is the research data collected by historians, based on written and cartographic sources and enriched with authority data. However, the potential of GIS cannot easily be used to evaluate this source base, as geodata can only be used to analyse topography and morphology, but not the functionality and temporal development of historical urban spaces (Kuhn et al., 2014; Purves et al., 2019). To achieve this, an ontology is needed to semanticise and contextualise the data. This geosemantic contextualisation allows geodata on the one hand and complex historical context information (semanticised historical location data) on the other hand to be queried and analysed together. To this end, the project develops an ontology that is partly based on the Historical Ontology of Urban Space (HOUSe), but fundamentally restructures and extends it. For this purpose, we use a modular ontology modelling approach (Shimizu et al., 2020; Hitzler/Krisnadhi, 2018).

Requirements analysis

The concepts of the HiSMaComp ontology are based on several pillars. In addition to the HOUSe ontology, there are numerous other glossaries and vocabularies - both digital and analogue - that are used as a basis for the ontology. To ensure good reusability and connectivity, the project's ontology is also linked to the top-level ontology CIDOC CRM and its extension CRMGeo.

In order to make concrete modelling decisions, the project carried out a comprehensive requirements analysis. This elementary component of modular ontology modelling is based on user stories and the formulation of competency questions.

However, the requirements analysis is not only aimed at the new development of the project's ontology, but also at the critical evaluation of existing ontologies in order to assess what can be reused and what adaptations and extensions are necessary.

In this way, we ensure that the project's ontology is optimally adapted to our requirements - the comparative analysis of urban space over time.

Structure of the ontology and data modelling

In light of these considerations, decisions pertaining to the modelling process are made, and the individual modules of the ontology are developed in an iterative manner.

The ontology is centred on the concept of events. With regard to the individual topographic objects (topoobjects) present within urban environments, a number of topographic events (topoevents) are

recorded. These events represent the lifespan of the object in question. The contents of the cadastral survey are recorded as a topoobject. It is also possible to make extrapolations and rewrites from this point in time. The constituent elements of the topoevents include the topographic type (topotype), the functions (e.g. administration, healthcare) and the geometry (in the form of a polygon). In order to facilitate the aforementioned geosemantic contextualisation, the entities that have been modelled in this way are integrated with the relevant geometries within the GIS, thus enabling their analysis as a unified entity.

Methodological criticism

The integration of GIS and ontologies presents a duality of opportunities and challenges, which, as with other digital methodologies, necessitate a critical and reflective approach (Dobson, 2019; Hiltmann et al., 2021). It is essential to acknowledge the epistemological constraints associated with our research data. Historical data, whether textual or cartographic, does not represent reality; rather, it is a construct that can be distorted by the subjective perspectives of those who created it and by the methods used (Drucker, 2011). However, our two approaches – GIS and ontologies – are based on different assumptions. For instance, uncertainty must be modelled in an entirely distinct manner in both approaches. Concurrently, both approaches provide disparate, in fact separate, perspectives for analysis. All of these aspects must be taken into account when formulating user stories and competency questions. Our objective is to achieve a balance between the expansion of our methodological spectrum and the management of the resulting complexity.

Outlook

The planned poster will provide an overview of the methodological model of the HiSMaComp project, subsequently focusing on one of its components, namely the linking of ontology and GIS. It will present the project ontology. Furthermore, the poster will address the requisite requirements analysis, method criticism and potential application scenarios, with illustrative examples. This section will elucidate the workflow of the HiSMaComp project, beginning with the research question and progressing to the utilisation of user stories and competency questions, the classes of the ontology, and an excerpt from the project's knowledge graphs.

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